



Smart Telescope Sort

User Manual · Version 1.2

macOS 15 or newer · Apple silicon and Intel · Free and open source

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Organize TIFF, FITS and JPG captures from several smart-telescope brands into one year / object library. The app does not connect to the telescope: you copy sessions onto your Mac first, choose that Captures folder, and the app recognizes the folder layout for you.



Smart Telescope Sort

Multi-brand Captures → Targets

HOW IT WORKS

- Copy the telescope's dated session folders into Captures. The layout is detected; Targets (year) folders are skipped.
- Pick a year, a month and the file types to move: TIFF, JPG/JPEG, FITS/FIT or All.
- Review file plan lists every file with its object, date and target folder.
- Optional Zip or Tarball backup. You name it, and a progress window shows it running.
- Sort moves the ticked file types into Targets (year)(object). A newer copy replaces an older one.
- Files already in Targets are marked Duplicate. You're asked Yes/No before they go to the Trash.
- Emptied capture folders are removed. A folder still holding unticked images asks Yes/No first.

Open user manual (PDF)

Source code on GitHub

Astronomy Observation Planner

Mac App Store

Smart Telescope Planner

iPhone & iPad App Store

Nothing moves until you press Sort eligible files.

CAPTURE LIBRARY

Ready to review

Dated session folders - detected

SOURCE CAPTURES

/Users/Shared/Astronomy/Captures

ORIGINAL TARGETS

/Users/Shared/Astronomy/Targets

PROCESSING TARGETS

/Users/Shared/Astronomy/Vespera Processing

BACKUP STORAGE

/Users/Shared/Astronomy/Sort Backup

DESTINATION

Choose a year and month

Objects decode into Targets (year)(DSO) for the detected capture layout.

FILES TO MOVE

☒ TIFF ☒ JPG / JPEG ☒ FITS / FIT ☐ All

BACKUP

Tarball (.tar.gz) → /Users/Shared/Astronomy/Sort Backup

Targets ... skipped as sources.

This session Change...

This session Change...

This session Change...

5 capture rows found

Destination shows where TIFF/FITS files will move

Capture folder	Year	Month	DSO object	Img folders	Files	Status	Formats	Destination
2026-09-10_22-15-03_observation_M31	2026	09	M31	1	3	3 new	FITS, TIFF	Targets 2026/M31
2026-09-12_01-40-22_observation_NG...	2026	09	NGC7023	2	3	3 new	FITS, TIFF	Targets 2026/NGC7023
2026-09-14_07-20-55_plan_Demo_Night	2026	09	M27	1	1	1 new	FITS	Targets 2026/M27
2026-09-14_07-20-55_plan_Demo_Night	2026	09	NGC7000	2	2	2 new	TIFF	Targets 2026/NGC7000
2026-09-18_23-05-11_observation_M31	2026	09	M31	1	2	2 new	TIFF	Targets 2026/M31

5 folders ready to inspect

11 image files to sort

Replace only when newer. Existing target files are kept unless the source is newer.

11 files eligible: 11 new moves, 0 newer replacements.

Review file plan

Sort eligible files

Preview ready for Dated session folders (TIFF, FITS / FIT) — no files have been moved.

The main window: the folder layout is detected, and every capture folder is listed with its object, file count and destination before anything moves.

1. Assumptions (read this first)

- **You already moved files off the scope** (USB, FTP, or Wi-Fi). The app never talks to Vaonis, Seestar, DWARF, or Origin hardware.
- **One brand per Captures folder.** The layout is detected from the folder you choose. Do not mix Seestar albums and Vaonis dated sessions in the same Captures folder.
- **You choose the file types:** TIFF, JPG / JPEG, FITS / FIT, or All. TIFF and FITS are checked by default; JPG is usually for viewing and sharing.
- **Apple Photos is not a source.** Export to a Captures folder if images only live in Photos.
- **Every brand uses the same library layout:** Targets {year}/{object}/ inside your Target Folder.
- **The whole Capture Folder is scanned,** every subfolder in one pass. Folders with no object name are listed together so you can name them all at once.
- **Year** comes from dated folder names when present; otherwise from the newest image file's modification date.
- **Object name** is decoded from observation / album / object folder names.
- **Nothing is overwritten.** If Targets already holds a different file with the same name, the new copy gets a number (IMG_0001 2 .jpg). Only byte-identical copies count as duplicates, including copies saved under a number.
- **Targets {year}** folders are never scanned as sources and are never removed.

1.1 First launch: terms and backup advice

The first time the app opens it shows the assumptions, the backup advice and the disclaimer: *BigSkyAstro is not responsible for the loss of data. We have built in many safeguards to prevent it. The user accepts all liability using this freeware.* Check **I have read and accept the above**, then press **I Understand**. Your agreement is saved once, in a locked PDF on your Mac, and is never overwritten; the next time the box is already checked and greyed out. **Help** → **Terms Acceptance Record...** shows when you agreed and where the record is kept.

1.2 What the app does not do

- It does not stack, stretch, or post-process images.
- It does not support Unistellar in this version.
- It does not invent missing object names — if a folder has no usable name, the object label may be the parent folder name.

2. Your three folders

The first time the app opens it asks where two library folders live. Check **Save this as default** to keep the choice; leave it unchecked to use it for this session only. Each folder has a **Change...** button in the main window, and a label beside it shows **Default** or **This session**.

- **Capture Folder** — where the images are located. Choose it with the **Choose...** button. The folder layout is detected each time the preview refreshes.
- **Target Folder** — where you want them put: it holds your Targets {year} folders. If it is not set, Targets folders go inside the Capture Folder.
- **Backup Storage** — where backups are written before a sort.

Status lights in the top right show each folder at a glance: **green** when it is selected (and, for the Capture Folder, available), **red** until it is chosen.

{Target Folder}/Targets {year}/{object}/
example: .../Targets/Targets 2026/M31/
example: .../Targets/Targets 2026/M42/

If the year folder does not exist yet, the app offers to create Targets {year} first. No files move when it is created.

3. Detected folder layouts

Telescope names identify folder layouts only. Smart Telescope Sort is an independent app and is not affiliated with, endorsed by, or created by those manufacturers. The detected layout is shown under the title, for example *Dated session folders · detected*.

Layout	Telescopes	What the folders look like
Dated session folders	Vespera, Stellina	2026-09-10_22-15-03_observation_M31
Object album folders	S30, S30 Pro, S50	M31/, NGC253/ (or a USB dump of albums)
Session folders	DWARF 3, II, mini	20260907_DWARF3_session/M33/
Object-and-date folders	Origin Mark II	M31_2026-09-05/

4. How each telescope gets files onto the Mac

Offload is done outside this app. Below: how vendors typically expose files and what the app expects after you copy them into Captures.

4.1 Vespera / Stellina

- **Off the scope:** FTP the **User/** dated folders. Newer models also support USB-C transfer. Multi-night: stop each night so the project saves, then copy each night's dated folder.
- **In Captures:** keep dated names unchanged, e.g. 2026-09-21_11-43-55_observation_M31 or ..._plan_My_plan with nested 01-observation-... and 01-images-....

```
User/      (on telescope)
  2026-09-10_22-15-03_observation_M31/
    01-images-initial/
      img-0001.tiff
      img-0002.fits
→ copy whole folder into Captures → Sort → Targets 2026/M31/
```

4.2 S30 / S30 Pro / S50

- **Off the scope:** USB cable (drive often shows **MyWorks** / object folders), Wi-Fi file share, or app export of FIT.
- **In Captures:** object albums with .fit / .fits inside (e.g. M31/, or a USB dump containing several albums).

```
MyWorks/
  M31/
    ...fit
  NGC253/
    ...fits
→ copy albums (or the dump) into Captures → Sort → Targets {year}/M31/
```

4.3 DWARF 3 / II / mini

- **Off the scope:** USB mass storage (appears as a disk) and/or FTP (often ftp://192.168.88.1 on the telescope's Wi-Fi).

- **In Captures:** keep each session folder intact. The object may be a nested folder (e.g. .../M33/) under a dated session.

```
Astronomy/<session>/
M33/
  stacked.fits
  stacked.tiff
  sub_0001.fits
```

→ copy session into Captures → Sort → Targets {year}/M33/

4.4 Origin Mark II

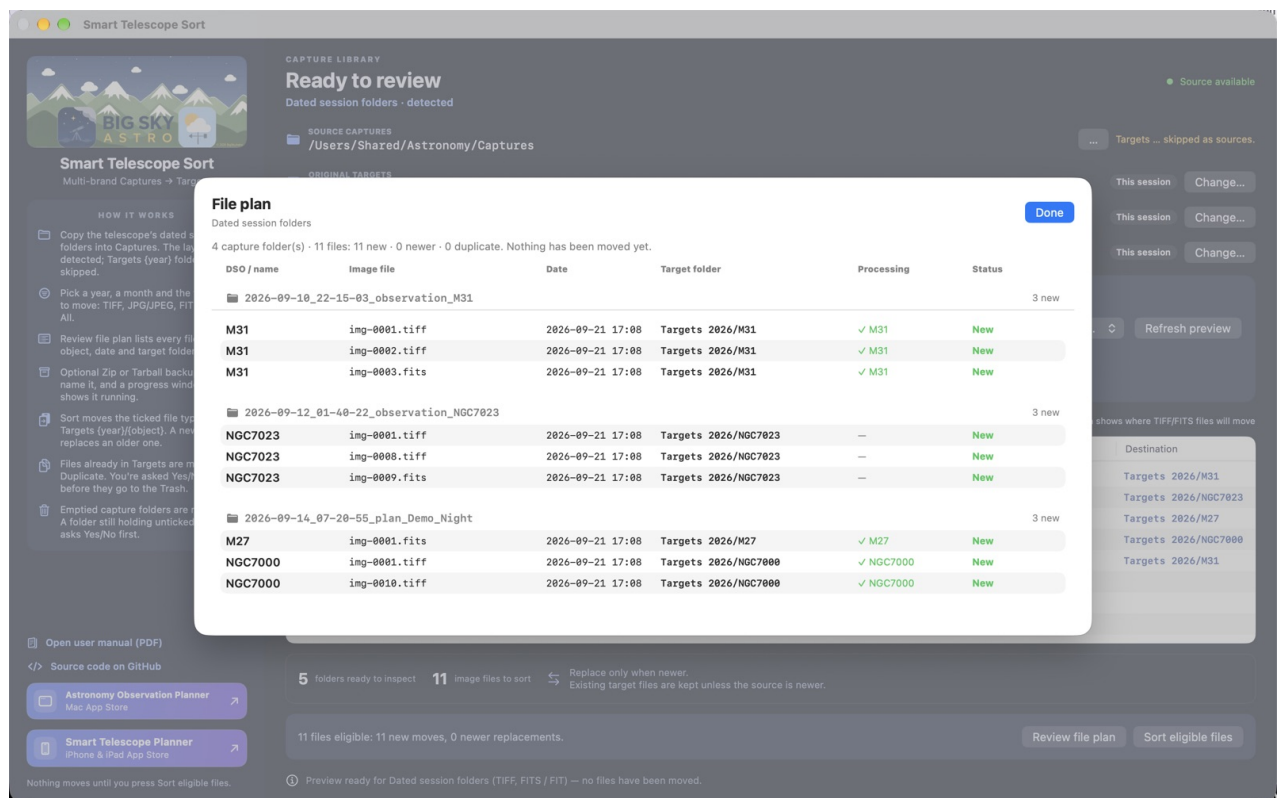
- **Off the scope:** enable **Save Raw Images**, then copy via USB stick (FAT32/exFAT) from the app File Manager, or FTP to a computer.
- **In Captures:** folders named object + date, e.g. M31_2026-09-05/ with .fits. A parent USB dump with several object + date folders also works.

```
M31_2026-09-05/
  raw_0001.fits
  raw_0002.fits
```

→ copy into Captures → Sort → Targets 2026/M31/

5. Step by step

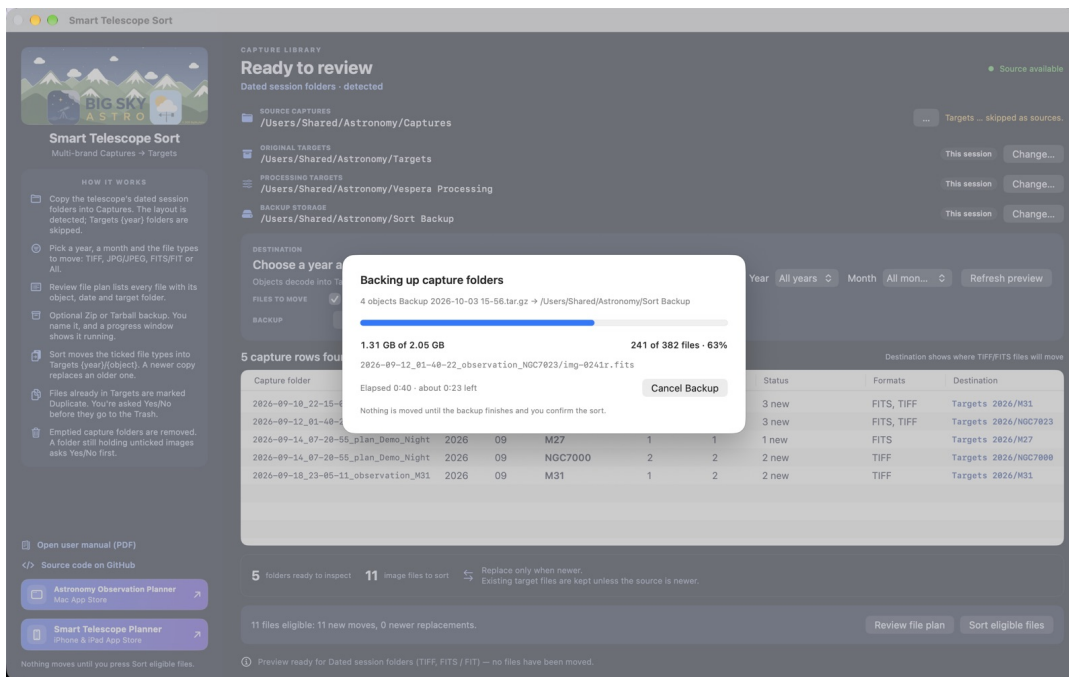
- Copy sessions off the telescope into a Captures folder.
- Open **Smart Telescope Sort** and answer the folder questions (first launch only).
- Press **Choose...** beside **Capture Folder** and pick that folder. The layout is detected and every subfolder is scanned.
- Pick a **Year** and **Month** (or All), and check the **Files to Move:** TIFF, JPG / JPEG, FITS / FIT, or All.
- Under **OK to Delete**, check **JSON** and **Astrometry** if those files may be deleted without asking (see section 7).
- Choose a **Backup:** Off, Zip (.zip), or Tarball (.tar.gz).
- Press **Review file plan** to inspect every file, its target folder and status.
- Press **Sort eligible files**. Name the backup, watch it run, then confirm **Sort now**.
- Watch the copy: each file is copied and checked (section 7). Answer the questions about plate-solve and calibration folders.
- When every move is done, say Yes twice to delete the originals, then Yes twice to delete the processed capture folders — or No to keep them.
- Browse Targets {year}/{object} for stacking and archive.



Review file plan lists every file grouped by capture folder, with its target folder and status. Nothing has moved yet.

6. Backups

- **Zip or Tarball** — chosen in the **Backup** menu. Before it starts you are asked to name it; the suggestion is the objects plus the date and time, e.g. M31 M27 Backup 2026-10-03 15-27.tar.gz. A number is added if the name is already used.
- A progress window shows the size and file count done, time elapsed and time left. **Cancel Backup** stops it and removes the partial archive; nothing is moved.
- When the backup finishes you are asked to confirm the sort. If it fails, nothing is moved.
- **Backup Off** — the app still offers **Back up first...**, which copies the capture folders into a named folder in Backup Storage, or **Sort without backup**.
- Archives are written fast (light compression) using the archiver built into macOS. If it is missing or a backup fails, **Help** → **Free Zip & Tarball Apps** lists free alternatives.



A named Tarball backup running, with Cancel Backup.

7. Sorting: copy, check, then delete

Sorting never moves a file in one step. Each file is **copied** into Targets {year}/{object}, the copy is **checked byte for byte** against the original, and only when every move is done are you asked about **deleting** the originals. A progress window shows the strategy, the file being copied, how many are done and the time elapsed; **Stop** ends the copy early and deletes nothing.

- **Plate solves** (astrometry folders, such as Vespera's 01-pointing-initial) are not sorted with the images. After the copy you are asked to **Move to Targets, Leave** or **Delete** them. Move puts them with their object: Targets {year}/{object}/Plate Solves/{session}. Sessions with no object name are listed together for you to name.
- **Calibration folders** (Lights, Darks, Flats, Bias, Master...) are offered the same way: **Move...** to a folder you choose, **Leave** or **Delete**.
- **Delete the originals** — once every move is done you are asked Yes / No, then **Are you sure?** Yes / No. No keeps them; the next scan marks them Duplicate.
- **Delete the processed capture folders** — the Capture Folder is scanned again and the app lists what is left in each processed folder (JSON, plate solves, other files). Again you are asked Yes / No twice. A folder still holding images not in Targets yet is never offered.
- Deleted files and folders go to the Trash where the drive supports it.

7.1 Duplicates

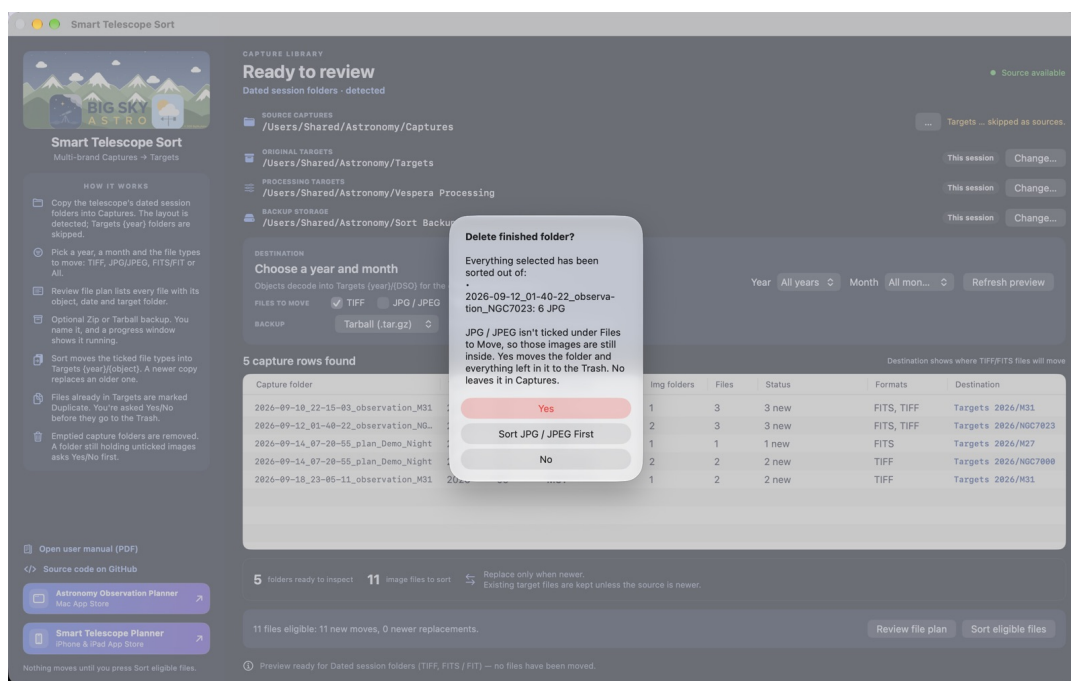
- **Duplicates** — files already in Targets, byte for byte, are marked *Duplicate*, even when the Targets copy was saved under a number such as IMG_0001 2 .jpg. **Delete duplicates...** asks Yes / No; the Targets copies are not touched.
- **Remove Identical Copies...** (beside the Target Folder) finds images saved more than once in a Targets {year}/{object} folder by earlier sorts. It lists them by object with the space they take, asks Yes / No twice, compares each copy byte for byte once more, and keeps one copy of each — the un-numbered one when there is one. Files that differ, files with other names, and Plate Solves are never touched.

7.2 JSON and astrometry files

- Telescopes leave .json session files and plate-solve (astrometry) files in the capture folders. Under **OK to Delete** in the main window, check **JSON** and **Astrometry** to let the app delete folders holding them without asking.
- Unchecked (the default), a pop-up lists how many of each a folder holds, with a checkbox for each. **Delete Checked** deletes only what you checked; **Keep All** keeps the folders.
- **Folders holding any other files** — Affinity Photo, XISF, Siril sequences, notes and so on — are never deleted by clean-up.

7.3 Finished folders

- When a capture folder is sorted but still holds images of a type you did not check (for example JPG), you are asked **Delete finished folder? Yes** moves the folder and everything left in it to the Trash, **Sort JPG / JPEG First** checks that type and sorts it, and **No** leaves the folder in Captures.



Delete finished folder? — Yes, sort the remaining type first, or No.

8. Safety rules

- Nothing moves until you confirm the sort.
- Copy, check, then delete: an original is deleted only after its copy is checked byte for byte and you say Yes twice.
- Nothing in Targets is overwritten; a clashing name gets a number.
- Targets {year} is never scanned as a capture source.
- The Capture Folder, the Target Folder, Backup Storage and Targets year folders are never removed during clean-up.
- Folders holding files other than images, JSON and plate solves are never deleted.
- Backups copy whole capture folders; they never modify Targets.
- Every deletion asks Yes / No first; deleting originals, processed folders and identical copies asks twice.

9. Free and open source

Smart Telescope Sort is free, and its source code is on GitHub: github.com/rderry/Smart-Telescope-Sort. Open it from **Source code on GitHub** (with the GitHub logo) in the sidebar or **Help** → **Source Code on GitHub**. The

app first asks for credit, then opens the repository when you click **Continue**.

If you change it and give it away, please give credit to BigSkyAstro: include the BigSkyAstro logo and a link to bigskyastro.com. Credit line: “Based on Smart Telescope Sort by BigSkyAstro — <https://bigskyastro.com>”.

10. Help and support

- Privacy policy: <https://bigskyastro.com/privacy>
- App page: <https://bigskyastro.com/macOS/smart-telescope-sort>
- Support: <https://bigskyastro.com/feedback/smart-telescope-sort> · support@bigskyastro.com

End of manual — Help → Smart Telescope Sort User Manual (Shift-Command-I) opens this PDF.